Questions

Q1.

Figure 8 shows a helium nucleus.

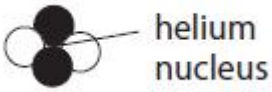

Figure 8

(i) Describe the difference between a fusion reaction and a fission reaction.

(2)

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(ii) Nuclear fusion does not happen at low temperatures because of electrostatic repulsion between

(1)

- ☐ **A** beta particles
- ☐ **B** electrons
- ☐ **C** neutrons
- ☐ **D** protons

(Total for question = 3 marks)

Q2.

(i) X-rays can be used in diagnosis and treatment from outside the body. Some x-rays are absorbed by bone as they travel through the body.

Figure 4 shows how the intensity of the x-ray beam gets less as the x-rays travel further through the bone.

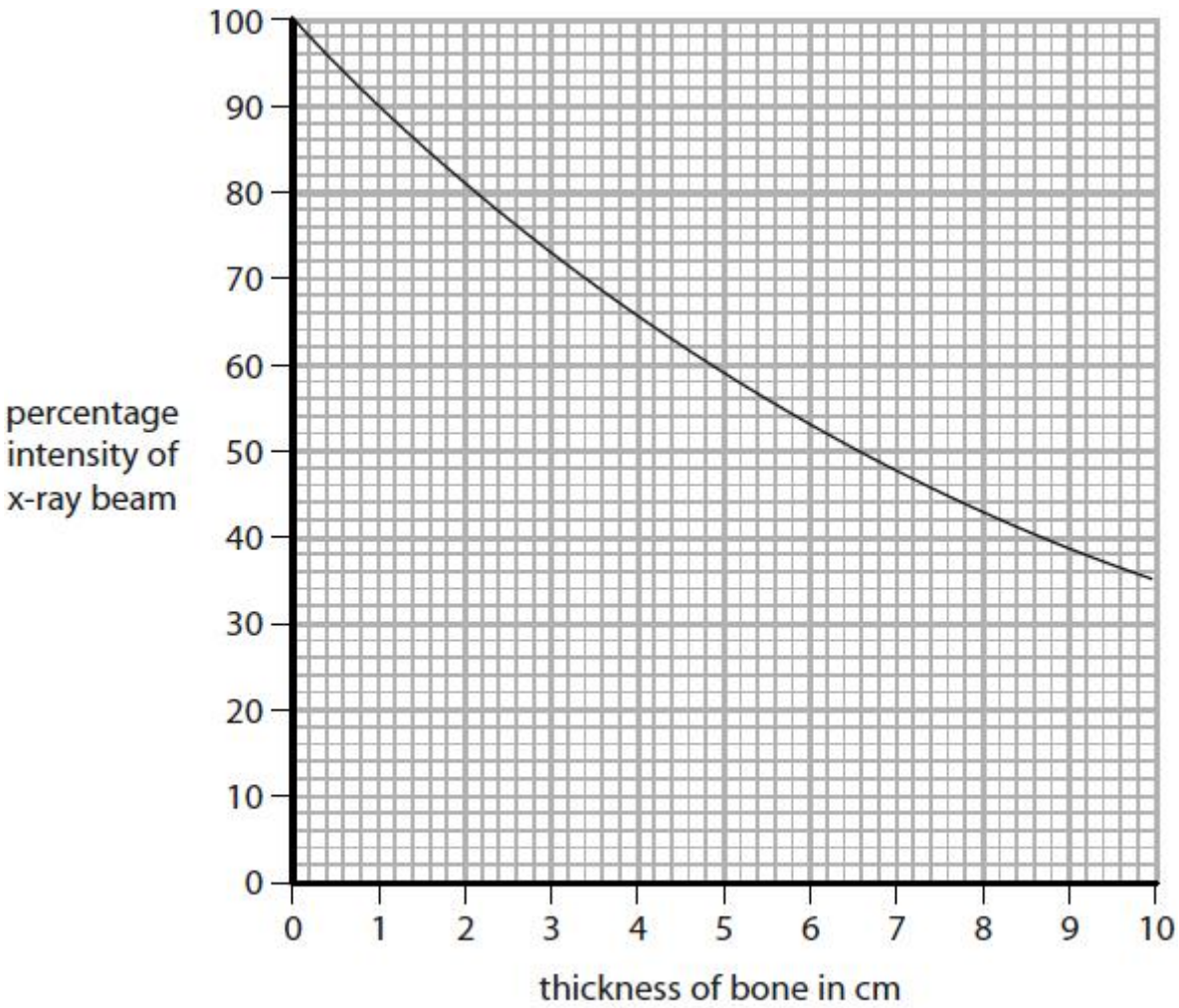

Figure 4

Use the graph to determine the thickness of bone that will reduce the percentage intensity of the x-ray beam by half.

(2)

thickness = ... cm

(ii) Radioactive isotopes may be placed inside the body for treatment.
The energy absorbed by tissue in the body needs to be known.

The number of joules of energy absorbed by each kilogram of tissue is measured in one of the units shown.

This unit is

(1)

☒ **A** kg/W

☒ **B** J/kg

☒ **C** kg/J

☐ **D** W/kg

(Total for question = 3 marks)

Q3.

An atom of beryllium has the symbol

$$^{9}_{4}Be$$

The beryllium atom has 4 protons.

(i) Determine the number of neutrons in this atom.

(1)

number of neutrons .....

(ii) Determine the number of electrons in this atom.

(1)

number of electrons .....

(Total for question = 2 marks)

Q4.

Carbon-14 is radioactive and has a half-life of 5 700 years.

The number of radioactive carbon-14 atoms in a very old piece of wood is found to have decreased from 1 000 000 to 125 000.

Determine the age of the piece of wood.

(2)

age of wood = ... years

(Total for question = 2 marks)

Q5.

Figure 8 shows two types of force that act in the Sun.

One type of force acts inwards, towards the centre of the Sun.

The other type of force acts outwards, away from the centre of the Sun.

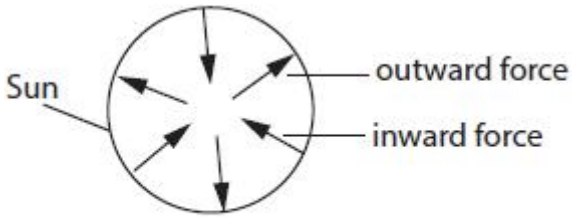

Figure 8

Nuclear fusion reactions take place in the Sun.

Describe what happens in a nuclear fusion reaction.

(2)

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(Total for question = 2 marks)

Q6.

This question is about nuclear reactions.

Nuclear fusion is a nuclear reaction that takes place in the Sun.

Describe what happens during a nuclear fusion reaction.

(2)

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(Total for question = 2 marks)

Q7.

Nuclear fusion provides the energy source for stars including the Sun.

Describe what happens during nuclear fusion.

(3)

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(Total for question = 3 marks)

Q8.

A nuclear fusion reaction is happening in our Sun.
The nuclear fusion reaction produces helium and radiates energy.

(i) State **one** of the conditions necessary for this nuclear fusion reaction to start.

(1)

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(ii) Describe what happens in this nuclear fusion reaction.

(2)

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(iii) The intensity of the Sun's radiation in W/m^2 on the surface of Earth is 1.32×10^3.
The intensity of the Sun's radiation in W/m^2 on the surface of Mars is 4.92×10^2 .
Calculate the ratio

$$\frac{\text{intensity of the Sun's radiation on the surface of Earth}}{\text{intensity of the Sun's radiation on the surface of Mars}}$$

(2)

ratio = ...

(Total for question = 5 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about radioactivity.

Alpha (α), beta (β) and gamma (γ) are three types of radioactive emissions.

Which statement describes **all** of these radioactive emissions?

(1)

☐ **A** ionising and emitted by stable nuclei

☐ **B** ionising and emitted by unstable nuclei

☐ **C** neutral and emitted by stable nuclei

☐ **D** neutral and emitted by unstable nuclei

(Total for question = 1 mark)

Q10.

Radioactivity from radioactive sources can be used in the diagnosis and treatment of cancer.

Describe **one** way radioactivity can be used in the diagnosis of cancer and **one** way radioactivity can be used in the treatment of cancer.

(6)

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(Total for question = 6 marks)

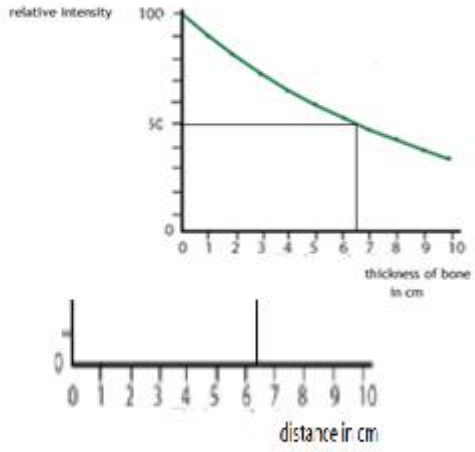
Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	a description referring to: fusion involves coming together / joining of particles / nuclei / atoms (1) fission involves (larger) particle(s) / nuclei /atoms breaking up (1)	 not just 'fuse together' that's just restating – more explanation needed particles etc. coming apart / separating no marks if just objects / things joining / coming apart	**(2)** AO 1 1

Question Number	Answer	Mark
(ii)	**D** protons **The only correct answer is D** *A 'beta particles' is incorrect, they are not found in nuclei to facilitate that repulsion* *B 'electrons' is incorrect, for the same reason as A* *C 'neutrons' is incorrect as they don't repel each other*	**(1)** AO 1 1

Q2.

Question Number	Answer	Additional guidance	Mark
(i)	Constructs a line across at an intensity of 50 (with a vertical to the thickness axis) (1) e.g.  (thickness =) 6.5 - 6.7 (cm) (1)	award full marks for the correct answer without working	(2)

Question Number	Answer	Mark
(ii)	**The only correct answer is B J/kg** None of the other options have units which are the same as J/kg	(1)

Q3.

Q4.

Question Number	Answer	Additional guidance	Mark
	processing (1) $\frac{125\ 000}{1\ 000\ 000}$ OR $\frac{1}{8}$ OR 3 half-lives or 3 x 5700 evaluation (1) 17 100	accept an appropriate attempt using more than one halving 17 000 award full marks for the correct answer without working	**(2)**

Q5.

Question number	Answer	Additional guidance	Mark
	description to include any **two** from: (two) isotopes/nuclei/atoms (1) fusing (1) release / emit energy (1) decrease in mass (1)	hydrogen joining / coming together allow heat for energy	**2** **AO1.1**

Q6.

Question number	Answer	Additional guidance	Mark
	A description including **one** from hydrogen nuclei/atoms join (1) helium is produced (1) **one** from lost (total) mass (1) mass is converted to energy (1)	nuclei/atoms join larger/heavier nucleus produced energy is released	**(2)** **AO1**

Q7.

Question number	Answer	Additional guidance
	A description to include any three of the following	
	(smaller) nuclei / atoms / particles (1)	two named eg hydrogen (nuclei)
	come together / join (1)	allow fuse not 'bond'
	to produce a larger nucleus / atom / particle (1)	helium for nucleus
	needing high temperature / pressure (1)	accept fast (moving) nuclei
	overcoming repulsion (between nuclei) (1)	
	energy released (1)	ignore energy created

Q8.

	Answer	Additional guidance	Mark
(i)	one from:		(1) AO1
	high temperature (1)	allow 'heat' for 'temperature' in this context	
	high pressure (1)		
	high (particle) density (1)		
	high (particle) speed / KE (1)		

	Answer	Additional guidance	Mark
(ii)	description to include: (two) isotopes/nuclei/atoms (1) fusing (1)	 hydrogen allow joining / coming together / bonding IGNORE collide	(2) AO2

	Answer	Additional guidance	Mark
(iii)	substitution (1) 1.32 (× 10^3) 4.92 (× 10^2) evaluation (1) 2.68	 accept 110 : 41 for 2 marks 11 : 41 for 1 mark POT error scores 1 award full marks for the correct answer without working	(2) AO2

Q9.

Question number	Answer	Mark
	B ionising and emitted by unstable nuclei A is incorrect stable nuclei do not give radioactive emissions C is incorrect not all radioactive emissions are neutral D is incorrect not all radioactive emissions are neutral	(1) AO1

Q10.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 **Diagnosis** Tracers radioactive sources swallowed by or injected into patients travel around the body emit gamma radiation (from radioactive decay) gamma radiation passes out through the body detected outside the body with a gamma camera PET Scanner (more of) the tracer goes to the cancer cells gamma radiation is emitted (gamma rays) from/in different directions gamma detected by gamma cameras (3D) pictures produced on a computer screen Named tracer, e.g. technetium, iodine, fluorine	**(6)** **AO1**
	Treatment Radiotherapy Use of gamma rays/x-rays kills cancer (cells) radiating small area of body (with gamma rays/X rays) radiation used for a short time (gamma rays) from/in different directions brachytherapy radioactive sources inside the body may not be removed put close to cancerous tumour **Ignore references to chemotherapy**	

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents a description with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents a description that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about cancer diagnosis or cancer treatment using radioactivity OR limited description about cancer diagnosis **or** limited description about cancer treatment using radioactivity OR allow isolated facts about cancer diagnosis using x-rays, C(A)T scans, MRI, ultrasound for level 1 if no other mark scored	Possible candidate responses Uses/emits gamma (rays) can kill cancer(ous) cells radiotherapy Cancer is treated using gamma rays CT scan is used to find tumours
Level 2	3–4	Additional guidance limited description about cancer diagnosis **and** limited description about cancer treatment using radioactivity OR detailed description of either cancer diagnosis or treatment using radioactivity	Possible candidate responses PET scans are used in diagnosis and radiotherapy is used to treat cancer OR Cancer is treated using gamma rays, the gamma rays kill the cancer cells

Level 3	5–6	Additional guidance detailed description about either diagnosis OR cancer treatment using radioactivity AND limited description about either cancer treatment OR diagnosis using radioactivity	Possible candidate responses Tracers are used in diagnosis and radiotherapy uses gamma rays to kill cells. OR Tracers emit gamma rays which pass out through the body and radiotherapy uses x-rays